Education and Information Technologies (2023) 28:5105–5145
https://doi.org/10.1007/s10639-022-11356-2

An effective deep learning pipeline for improved question classification into bloom's taxonomy's domains

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Received: 7 May 2022 / Accepted: 13 September 2022 / Published online: 28 October 2022
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Abstract

Examination assessments undertaken by educational institutions are pivotal since it is one of the fundamental steps to determining students' understanding and achievements for a distinct subject or course. Questions must be framed on the topics to meet the learning objectives and assess the student's capability in a particular subject. The generation of examination questions from extensive text material is challenging and complicated. For example, massive volumes of textbooks make it time-consuming for faculties to annotate good-quality questions, keeping them manually well balanced. Thus, teachers rely on the Bloom's taxonomy's cognitive domain, a popular framework to assess students' intellectual abilities. This study's motivation is to propose a pipeline that could provide new questions from a given text corpus that could be retrieved from a particular input. These generated questions could be incorporated into a question recommender while being automatically classified under the specific cognitive domain under the Bloom's taxonomy. Literature reviews showed that the work done over the Bloom's taxonomy domain had obtained results by implementing classical machine learning methods and few with deep neural networks. The proposed network architectures have shown remarkable results and state-of-the-art architectures compared to the literature. This research study concluded that the pipeline is effective and significant in generating questions, like manually drafting questions, categorizing them into the Bloom's taxonomy's domains, and providing explicit content-based question recommendations.

Keywords Bloom's taxonomy · Content-based recommendations · Deep learning · Natural language processing · Siamese neural networks · Transformers

+ Harsh Sharma, Rohan Mathur and Tejas Chintala, contributions are equal.

1 Introduction

The key objective of teachers in the teaching domain is to provide a better way of studying and obtain a suitable medium of assessments to measure and analyze students' cognitive levels. Therefore, the written exam is a mode for various faculties to uphold a student's knowledge and understanding and assess students' learning in each situation. Contrastingly, the current framework of the question annotation requires manual rigor, time, and challenging work since they are supposed to be diverse and robust so that they can be utilized to assess a student's thinking ability and exploration of a particular topic or skill set (Alstete & Beutell, 2019; Cormack et al., 2020).

The Bloom's taxonomy (Bloom, 1956) is considered an efficiently proven and accepted framework for educational assessments. Teachers commonly use this framework to create evaluations that target methods like project-based learning and simplifying an activity to help personalize learning. It consists of three diverse learning domains: cognitive, affective, and psychomotor (El Asame et al., 2022; Caprara & Caprara, 2022). Each of these domains consists of a hierarchy that signifies various levels of learning. It is important to note that the various levels of assessment with this taxonomy are particularly hierarchical, i.e., the higher levels of the given taxonomy depend firstly on obtaining the prerequisite knowledge and skills at lower levels. If a particular taxonomy level is to be analyzed, the level below it has already been analyzed by hierarchy follow-up. Although many versions of the Bloom's taxonomy have come up in recent years, the first proposed framework in 1956 has been used.

Adaptive E-Learning has gained popularity since it aims to increase effectiveness as part of monitoring learning analytics according to the Bloom's taxonomy (Hubalovsky et al., 2019). Question generation is a core aspect and one of the most dominant Natural Language Processing (NLP) tasks. Research has been conducted on question generation incorporating various studies like reading comprehension (Heilman & Smith, 2010; Indurthi et al., 2017) and community-based Question answering methods (Qiu & Huang, 2015; Quan et al., 2018; Chen & Li, 2020) implemented in Quora and StackOverflow. Numerous methodologies have been implemented for generating questions. Such approaches propose different directions to deploy it on a research background for educational purposes (Le et al., 2014). Many state-of-the-art methods and models have been applied to perform the question generation tasks - like the Bidirectional Encoder Representations from Transformers (BERT) and T5 transformer (Singh & Sharma, 2018; Devlin et al., 2019; Raffel et al., 2020). Question classification can be defined as selecting a specific category for a given question from a predetermined list of categories based on contextual information (Stanescu et al., 2008; Ray et al., 2010; Poorman & Mastorovich, 2020).

There is a significant difference between general text classification and question classification, especially in the defined context. Regarding classifying the questions, there is an issue with handling short sentences, unlike text documents, which have comparatively more content (Samiappan & Chakrapani, 2013; Wasim et al., 2019; Yeoh et al., 2021; Singh et al., 2021; Wijanarko et al., 2021). The sub-part of the Bloom's taxonomy, i.e., its cognitive level, has emerged as a highlight or blooming topic in research and a fascinating field to explore its associated aspects like

understanding and assessing students. Various frameworks and algorithms have been experimented with and implemented for Bloom level-based classification of levels, such as unsupervised techniques for semantic space clustering like Latent Dirichlet Allocation and supervised classical machine learning algorithms. However, recent works that worked on this specific domain showed implementing different statistical methods like Term-Frequency inverse document frequency (TF-IDF) and N-gram, along with Support Vector Machines (SVM) and K-Nearest Neighbours (KNN) based algorithms. There is a shortage of techniques for categorizing questions by including the use-case of deep learning and state-of-the-art architectures and implementing standard or contextual word embeddings.

On the contrary, framing diverse questions for a particular topic is vital. This could be exhaustive whenever a faculty or teacher wants to get contextually diverse questions from the same topic. Manually drafting or annotating questions for a given topic is crucial. It is supposed to explore and assess a student's thinking ability. Such manual drafting over 'n', several topics would require much time and effort, especially if the corpus of a particular subject is immense. This concern demands a well-founded solution that could save time and provide a proficient level of diverse questions with little or no extra effort. Therefore, to overcome the issue of question generation by teachers and save considerable time and effort (Hudson & Moubayed, 2021; Palivela, 2021; Priya et al., 2022). This study showcases a suitable pipeline for effective question generation using state-of-the-art architecture. An effective supervised model is dedicated to classifying questions for categorizing the Bloom's taxonomy's specific cognitive domain and providing explicit recommendations based on semantic similarity distance between generated questions and manually drafted questions controlled via a certain input threshold (Ramnarain-Seetohul et al., 2022). The threshold value can range from 0 to 1. It can be set to more than 0.5 to provide the most general recommendations. However, one can control the threshold value according to the obtained similarity score from the questions. Thus, the similarity scores are being retrieved.

This research work proposes the following pipeline:

- Generation of questions using state-of-the-art transformers of good semantic quality is helpful for basic question preparation tasks in education.
- Categorization of questions on the various levels of the Bloom's taxonomy, specifically the cognitive domain.
- Using a Long Short-Term Memory (LSTM) based Siamese NN model (Jonas & Aditya, 2016) to compare the semantic similarity between the model-generated and manually-drafted questions and provide the diversity between them.
- Explicit question recommendations based on semantic similarity distance obtained from an LSTM-based Siamese NNs between generated and manually drafted questions. A cosine similarity controlled over distance metric is used through a certain input threshold.

1.1 Contributions of the study

The present research aims to create an effective, generalized, end-to-end pipeline for generating questions and classifying these questions based on the cognitive level, a subdomain of the Bloom's taxonomy and provides explicit recommendations for context-based questions. The various modules of the proposed pipeline have demonstrated the ability to provide generic responses to the input context. Thus, a generation module displayed a wide range of questions with reasonably good results; the recommendation module generated explicit questions based on the given context controlled by the input threshold.

Different datasets were considered for each subcomponent of the proposed pipeline to examine the strength of the proposed model. They were used in training and testing to check the pipeline's ability to provide a generic response.

The proposed pipeline design effectively addresses the difficulty of question generation and classification according to different levels of cognitive domains of the Bloom taxonomy, along with detailed question recommendations. This end-to-end pipeline receives a variety of questions directly from faculty by entering a text corpus once, receives the generated questions directly under the Bloom taxonomy in the cognitive domain, and evaluates the institution's assessment level. This complete pipeline promises to save additional time, rigor, and effort while providing a comprehensive solution. Although the focus of this work was the original version, one limitation is that the cognitive domains of these versions require experimentation. This work examines and experiments with different contextual passages to generate questions. This proposed pipeline is useful for extracting contextual information using sequence-to-sequence learning.

2 Literature review

Recent research works and findings were explored concerning question generation, Question Classification, computing text similarity, the Bloom's taxonomy, and associated hierarchies using NLP.

2.1 Bloom's taxonomy-based applications

Harmon et al., (2010) discussed the challenges and shortcomings of the architecture of online-based assessments. They showcased statistically how often the implementation of dedicated proctoring affects a student's choice to manipulate. This work suggested altering the tactics to prevent reusing questions and presenting them. They were syntactically different by using a pool of questions (comprising different question sets) and finding ways to generate high-caliber questions. Radmehr & Drake (2018) proposed an assessment-based model that can be used to review teaching or learning-based assessment of mathematics subjects, exemplified in the case of integral calculus. This study promised to use two specific frameworks - Revised Bloom's taxonomy and metacognition framework to formulate and plan questions for taxonomy's knowledge types and cognitive processes.

Bogdanova & Snoeck (2019) developed an effective conceptual data modeling educational-oriented platform concerning the Bloom's taxonomy for providing necessary examples of various levels with dedicated learning results. Chilukuri (2020) proposed a framework for course design using the course's associated elements mapped to basic levels of outcome defined by the Bloom's taxonomy concerning learning outcomes. This work emphasized various techniques like interactive videos, learning distinct levels of knowledge, reciprocal presentations, and analytical and application-level learning. Saadullah & Elsayed (2020) showcased an audit simulation of different procedures in the revenue process, providing different methods for students to learn. This method was focused on teaching cases for the Bloom's taxonomy. Waite et al., (2020) proposed a revised framework for the Bloom's taxonomy, dedicated explicitly to mentorship provision for senior and junior faculties, which could help in career progressions. They used knowledge and cognitive level in their proposed framework.

2.2 Question generation

Lindberg et al., (2013) worked on a semantic-based method to obtain information and generate relevant and appropriate questions for an online-based learning structure. They proposed a novel, template-based framework for automated learning techniques that handpicks questions that show potential learning value. Questions generated through this are framed so that answers cannot be directly distinguished from the provided sentence/passage. For questions with learning value, their classifier received precision and recall scores of 0.47 and 0.22, respectively, with scores of 0.79 and 0.92 for questions with no learning value. This method of generating questions goes along with the categorization model devised using the Bloom's taxonomy method, i.e., to have a generated pair of questions that meet all the given cognitive levels of learning. Aithal et al., (2021) worked on Question answering to generate the possible question-answer pairs, given a passage automatically. Using pre-trained models, they have carried it out on "question"-"answer" pairs obtained from the Stanford question answering dataset (SQuAD). They also devised a question similarity procedure to isolate irrelevant generated questions.

2.3 Question classification

For Question Classification, various approaches were implemented and considered in the computer science domain, especially machine learning. Van Hoeij et al., (2004) showcased a model developed for the Bloom's taxonomy classification, specifically for its cognitive domain to interpret and evaluate short essay questions. Their work compared cognitive levels assigned by raters between subject matter experts and non-subject matter experts' groups. However, given the short text length, like essays and more instructions, this approach might still require more data and an improved algorithm for classification to be considered for application in a real-world scenario. Chang & Chung (2009) classified the Bloom's taxonomy to assess and classify English subject questions, including 14 general keywords for the Bloom's taxonomy and 288 test items. Their computed analysis showcased that taxonomy's "knowledge"

level has a 75% correct match. This evaluation proved that their work also needed some improvement. Omar et al., (2012) showcased and devised an automated analysis-based framework for given examination questions for diverse the Bloom's taxonomy categories with a rule-based approach to identify crucial keywords and verbs. They were explicitly focused on the computer programming domain and trained and tested it with about 100 questions. Haris & Omar (2015) also proposed a rule-based approach and N-gram, which improved the classification task. A dataset considered for the work, including around 135 questions initially in a computer programming domain in the English language, was further split into training data of around 100 and testing data of around 35, respectively. The classification was done between three different given diverse topic areas: Object-Oriented Programming domain, Software Engineering domain, and Algorithm and Data Structure. They were able to achieve an 86% of average F1 score.

Mohasseb et al., (2018) study showed various proposed approaches for question categorization and relative classification using a grammar-based approach rule. Their work identified different patterns amongst classification tasks and included implementing various machine learning algorithms to classify them by exploiting the given structure of those questions. Muhammad et al. (2019) proposed developing a multi-label corpus for biomedical questions classification focused on retrieving semantic features for classification. They had a unique approach for classifying biomedical questions and a novel methodology for lexical answer type prediction using a label power set. Mohammed & Omar (2020) showcased the implementation of modified TF-IDF. They extracted word2vec from a given corpus for developing a model to implement multi-class classification on specific Bloom's taxonomy's cognitive domain. Their work consisted of two datasets. One was a dataset from various sources like books, websites, and previous works. The other dataset has around 600 questions on diverse subjects (Yahya et al., 2012). Pre-processing included tokenization, stemming, and part of speech tagging along with word2vec for feature extraction. They have implemented SVM and KNN for performing classification and achieved 71% and 83%, respectively.

Boussakssou et al., (2020) proposed a reinforcement learning-based adaptive e-learning system that could generate adaptive paths per the user's dedicated learning path chosen. This work showcased challenges faced by the online learning system and different approaches to improve it using a dynamic approach and instructional design strategies. Using BERT's state-of-the-art architecture model, Geetha & Renuka (2021) showcased methods for improving sentiment analysis's given aspect for multi-label classification tasks. They implemented and compared BERT uncased model's performance on the product reviews against Naive Bayes, SVM, and LSTM.

2.4 Computing text similarity

Computation of input text similarity is one of the vital principles for determining closeness between two raw text inputs. This technique could be leveraged to derive the deep semantic relations and the similarity between text inputs at a specific level. Takano & Kajikawa (2019) explored the research area of technologies related to the Internet of things by extracting information on commercialization opportunities by

including the use-case of measuring text similarity between original papers and patents. They achieved satisfactory results using linkage analysis and cluster classification using cosine similarity. Nguyen et al., (2019) presented a novel method based on interdependent representations of short texts for obtaining the degree of semantic similarity by representing them as two dense vectors. The first dense vector is made up of the word-to-word similarity obtained using a pair of pre-trained word vectors; the second dense vector is built upon the word-to-word similarity based on outside sources of understanding.

Yang et al., (2021) suggested an approach for measuring text similarity, especially for short text, by combining semantic and syntactic information. The study used similar terms to construct semantic representations of smaller text sizes and utilized a parse tree to capture that syntactic similarity. Saedi & Dras (2021) showcased authorship identification on many datasets with vast numbers of authors, looking at various energy functions using deep learning based Siamese NN architectures. Their approach claims to identify texts previously unseen by authors and to surpass previous approaches. Olson et al., (2021) proposed novel methods for yelp reviews-based analysis as a collective representation of a city and its neighborhoods via a deep autoencoder-based approach and 1024 dimensions of deep text embeddings with an accuracy of around 77%. Marina et al., (2021) proposed a Siamese NN to detect and recommend complementary products. They used LSTM-based Siamese NNs, and a convolutional neural network (CNN) based Siamese NN. They compared the respective performances for semantic computing a relationship between product pairs. Later, LSTM-based Siamese NNs were used as an add-on. They made cosine similarity computations to detect relationships between complementary products using the main product titles.

Based on the literature review, the unique features of the pipelines have been analyzed. Question answering is to derive new potential question-answer sets on provided a passage (Aithal et al., 2021). The drawback faced when a few system-generated questions according to the passage's context was not understandable and irrelevant. More challenges were faced with short-length raw datasets and targeted keywords, which lacked contextual information within the text corpus (Van Hoeij et al., 2004; Chang & Chung, 2009). The researchers are motivated to consider diverse categories and domains of text corpus while devising the end-to-end pipeline for the current research study as the outputs (Omar et al., 2012; Haris & Omar, 2015; Mohasseb et al., 2018). The results generated should be generic as per user's input. The obstacles the online learning system faces in public demand the need for state-of-the-art architecture implementations (Boussakssou et al., 2020). Thus, the literature reinstates that more work is needed to improve the dataset and model's performance. Hence, this study focuses on forging a relevant pipeline for deep learning-based question generation and Question Classification, along with computation of similarity between the generated questions and the manually annotated ones with explicit content-based question recommendations. Deep learning-based networks and state-of-the-art architectures were experimented with and proposed as the final pipeline for better performance than existing models to achieve better results and efficiency.

3 Proposed pipeline for question generation with automatic bloom's taxonomy classifier and explicit recommendations

The primary purpose of this research is to devise an effective pipeline that could generate the questions obtained from a compilation of texts (Lu et al., 2021; Barbhuiya et al., 2021; AlArfaj & Mahmoud, 2022; Blšták & Rozinajová, 2022; Zhang et al., 2022). Then, these generated questions could be automatically classified under Bloom's cognitive domain. Another important sub-component of the pipeline includes scanning given questions to check for similarities among the given questions. After computing an evaluated similarity and its uniqueness, the proposed module can recommend questions built upon the input's context for a given metric evaluated via the architecture of devised deep NNs (Balaha & Saafan, 2021; Ramos et al., 2021). The first proposed research pipeline consists of a Question Generation Module for generating questions using a state-of-the-art transformer. A classification module for categorizing the questions assumed from Bloom's cognitive level was developed using specific deep learning based NN architecture. Apart from classifying these questions, the given pipeline also consists of another pipeline that evaluates two given questions. This module generates one Question, and another is provided as manually annotated. Based on the Manhattan distance and given input threshold, provide content-based recommendations of context-based questions via cosine similarity Siamese-based deep NN. Figure 1 (a) and (b) describe the proposed pipeline in the present work.

The proposed architecture can take multiple texts and perform the pre-processing of the text corpus. The T5 model was fine-tuned using the PyTorch lightning library. After fine-tuning, the text corpus was passed to obtain the generated questions. Generated questions could pass through the pre-processing module and an embedding layer to extract the featured vector representations, passing through a dense NN to classify Bloom's cognitive domain. Once generated questions are obtained, the end-user could get explicit recommendations of questions stored in a database, as per computed similarity distance metric between generated and manually drafted questions, controlled over a certain threshold. However, the user can also retrieve the questions' recommendations with custom text inputs, bypassing the pipeline.

4 Question generation module

4.1 Dataset preparation

The dataset used for question generation for the proposed framework is SQuAD 2.0 Dataset. SQuAD 2.0 is the extended version of the original version of SQuAD V1.1 (Rajpurkar et al., 2016, 2018; Zhou et al., 2017; Khodeir et al., 2018; Das et al., 2019; Lang et al., 2021; Chatzikonstantinou et al., 2021). SQuAD is a complete reading comprehension dataset, including questions and answers with the given context and where the answer is specified within the comprehension. This data has been picked up from multiple Wikipedia articles, and the curated questions are by crowd work-

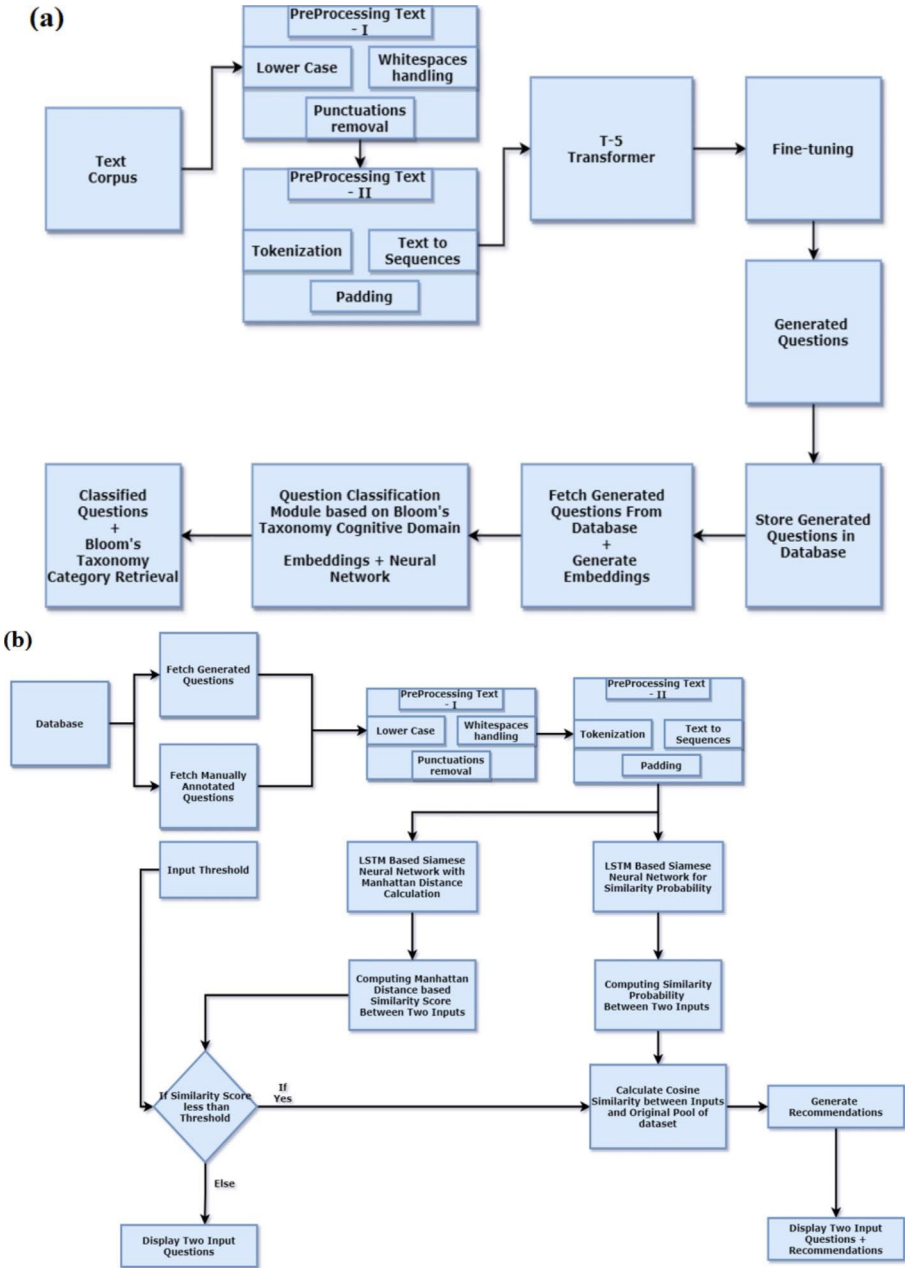


Fig. 1 **(a).** Joint pipeline for question classification with transformers and structured question classification with deep NN, **(b).** Joint pipeline for computing semantic distance linking generated questions and manually drafted questions and providing explicit recommendations of questions based on input threshold

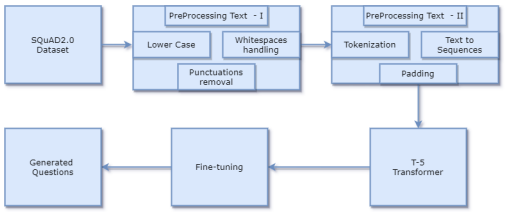


Fig. 2 The proposed pipeline for question generation

ers and are of a high standard. The original dataset (SQuAD 1.1) consisted of over 107,000 questions with their respective answers.

4.2 Preparing SQuAD dataset

The dataset has been made into an NLP format with columns (context, answer, question). The data consists of 86,821 rows of triplets.

- **Context**: The passage containing the answers to the questions. It is usually five sentences long.
- **Question**: The Question that is being framed, keeping context in mind.
- **Answer**: The answer to the corresponding question.

4.3 T-5 transformer

T5 transformer is an encoder-decoder-based model that inputs NLP-driven problems and converts them to a text-to-text specific format using the teacher forcing method. Training a T5 model requires an initial input sequence and a target sequence. It represents the latest research from the last couple of years on the same basic idea: training large NN on massive amounts of unlabeled text, then fine-tuning them on labeled text for performing tasks per user.

4.3.1 Pipeline of question generation

The current transformer was fine-tuned on a pre-trained T5 model called - "t5-base". Models like T5 and BERT have come up with various tokenizers; this current work uses the tokenizers from "t5-base". The T5 tokenizer gives us the respective input ids and attention masks for a sentence. The input ids convert the sentences into unique tokens, and the attention masks signify the presence of the specific tokens using '0' and '1' where 1 signifies the presence of the token. The sentences are also encoded to the maximum length. The entire pipeline is shown in Fig. 2. For T5 training, PyTorch lightning, a wrapper on PyTorch, was implemented to simplify the training process. The T5 tuner is a PyTorch lightning wrapper-based class containing essential helper functions like data loaders, forward, training, and validation step functions. This class was also used to fine-tune the pre-trained model. The fine-tuning included adding the input and output text (the questions) into one corpus and the target text, which are the questions for that answer.

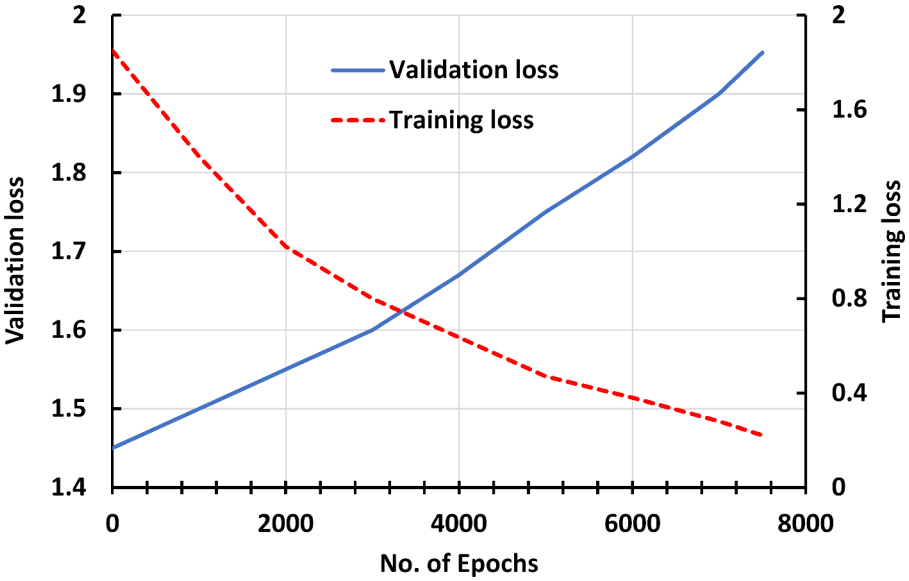


Fig. 3 Training and validation losses vs. No. of Epochs

After converting the sentences to tokens, the given training data was shuffled. The first 3000 rows were extracted for training, and padding of the given input data was done, with context having a maximum length of 512 and for the given output data (the generated Question) a maximum length of 96. Laboratories are calculated to find the loss for the required tokens. The input ids and its input attention mask are found for each triplet, along with the output ids and its output attention mask. The PyTorch lightning module inherits the t5.finetuner class. The forward function takes four values within this class (input id, input id attention mask, output id attention mask, labels). The output id is not needed here; it is being taken using the labels. The Adam optimizer is being used, giving the learning rate as "0.05" and with a given batch size of 2 for 5 epochs.

Parameters for fine-tuning the SQuAD Dataset: Batch size=2; Epochs=5; Learning rate=0.05; Optimzer=Adam; Dropout=0.0). After training, a steady decrease was observed in training loss, which was 0.261, as observed in Fig. 3. The validation loss, which turned out to be 2.16, fluctuated during training and has been depicted in Fig. 3. A key point to note here is that training could be done for only a stipulated amount of time due to the limited consumption of computation resources. Therefore, the fluctuations were recorded in validation loss. A sample of a generated question is provided in Table 1.

Table 1 A Generated Question, Given an Input Context and answer

Context	Answer	Generated Question
Proteins are the most abundant biomolecules of the living system. Chief sources of proteins are milk, cheese, pulses, peanuts, fish, meat, etc. They occur in every part of the body and form the fundamental basis of the structure and functions of life. They are also required for the growth and maintenance of the body. The word protein is derived from the Greek word "proteios" meaning primary or prime importance.	greek word originate	From what Greek word does "proteios" originate?

Table 2 Frequency Distribution of Different Datasets used for Classification of Questions

Cognitive domain	Yahya et al., (2012)	Mohammed & Omar 2020	TREC training set five datasets	Collected dataset
Knowledge level	100	26	26	20
Comprehension level	100	23	26	20
Application-level	100	15	26	20
Analysis level	100	23	27	20
Synthesis level	100	30	27	20
Evaluation level	100	24	27	20
Total	600	141	159	120

5 Question classification module

The authors propose a deep learning-based method for multi-class classification of Bloom's cognitive level, which takes an input text and attempts to classify it from given categories. The model was designed to get the best generalization accuracy.

5.1 Dataset preparation

Various datasets were considered for implementing multi-class classification specifically to the Bloom's taxonomy's cognitive domain, and some were manually collected. The initial dataset considered was given by Yahya et al., (2012), which consisted of over 600 questions, covering a wide variety of questions on different subjects (Mohammed and Omar 2020). This dataset had 141 open-ended questions belonging to the cognitive domain. The third dataset was obtained from some proportion of the Text REtrieval Conference (TREC) Question Classification dataset. The fourth dataset was collected manually from books and websites. The TREC dataset consists of diverse collections of sub-datasets of Question Classification consisting of open-domain, fact-based questions divided into various categories. It consists of various training and test datasets with different amounts of labeled questions. The one considered for this work was the TREC - Training set 5, which consists of 5500 questions. The dataset was labeled six classes per 6 different cognitive domain levels. Table 2 shows the question distribution at each level in the datasets. Tables 3 and 4 show a portion from both datasets.

Table 3 Sample of datasets used for question classification training – I

Dataset	Yahya et al., (2012)		Mohammed and Omar (2020)	
Cognitive domain	**Questions**	**Area**	**Questions**	**Area**
Knowledge level	How many years does this animal live?	Biology	Define morphology	Biology
	How much data can be stored in the cache memory	Computer science	List valuable information you can remember from the lesson.	Literature
Comprehension level	Describe four types of coupling in software design	Computer science	Describe how interest rates affect the economy	Economy
	Describe how Phillip and Timothy survived on the Cay	English literature	What is the difference between atomic and molecular mass?	Chemistry
Application level	Calculate for x in the following equation: 12x = 36	Mathematics	Sketch an experiment to see how plants grow in diverse kinds of soil.	Biology
	Derive a kinetic model from experimental data	Chemistry	Write the definition of displaySize() method for each class.	Computer science
Analysis level	Analyze the characteristics of frogs	Biology	Discriminate the distinct parts of the story (introduction, development, climax, resolution.)	Literature
	Contrast building in the coastal zone with building in a river floodplain	Geography	Analyze a work of art in terms of form, color, and texture.	Art
Synthesis level	Synthesis Level Compose an original work that incorporates five common materials in sculpture	Art	Write a simple JAVA program to devise the Inheritance concept	Computer programming
	Create a new product. Please give it a name and plan a marketing campaign	Marketing	Construct an alternative way to finish the story	Literature
Evaluation level	Assess the appropriateness of an author's conclusions based on the evidence given.	English literature	Outline how class ArrayList could be implemented using an array	Computer science
	Evaluate a work of art, giving the reasons for your evaluation	Art	What criticisms could you make?	Literature

The dataset consisted of several types of questions. Thus, it required some initial text pre-processing with the Natural Language Toolkit library, like removing punctuation symbols, handling white spaces, removing unnecessary symbols, and then exporting it to a data frame. Table 3 depicts a portion of the total questions of various categories in the given datasets. Tables 4 and 5 depict a portion of questions with their associated topic areas in different datasets.

Table 4 Sample of Datasets used for Question Classification Training – II.

TREC dataset & Collected Dataset		
Cognitive level	**Questions**	**Area**
Comprehension level	In your own words, tell what the story is about	English literature
	Illustrate a glycosidic and phosphodiester bond	Biology
Application level	Interpret the graph and state how many trees were cut down to produce paper	Mathematics
	How was Teddy Roosevelt related to FDR?	History
Analysis level	Can you explain why this is so essential in hydraulics	Physics
	Experiment with the verification of ohm's law	Physics
Synthesis level	Can you make a saline solution	Chemistry
	Find an unusual way to communicate the story of a book you have read.	English literature
Evaluation level	Do you agree with this proposal	Marketing
	Establish criteria for your improvement plan	Marketing

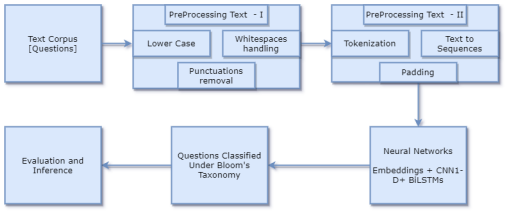


Fig. 4 Proposed sequential pipeline for question classification

5.2 Neural network pipeline for question classification for cognitive level of Bloom's

5.2.1 Taxonomy

The present work is applied to an ensemble and hence implements several types of combinations of NN architectures:

ConvNet1-D, LSTM, Bidirectional LSTM, BERT, Tensorflow's Embeddings Layer, ELMo Embeddings, and Embeddings were obtained using the GloVe algorithm. The general pipeline is depicted in Fig. 4.

5.2.2 Pre-processing phase

Before working on the core NN, all questions were pre-processed with the following steps:

- Converting them to lowercase and handling extra white spaces between them.
- Punctuation removal and unwanted symbols removal.
- Remove words whose length was less than 2 and more than 21 characters to keep the necessary and semantically meaningful words for training.
- Using the Scikit-learn package, an initial train test split (80: 20 ratio) was carried out, where 80% is the training data and 20% is for test data.

- Using the TensorFlow package, tokenization, text to sequences, and padding were done. This helped to convert the raw text to tokens to get indexes of unique words and build word vocabulary for training and thus integer type sequences of words so that they can be passed through the network with the following parameters:
- The vocab size = 1692, max length = 40.
- After converting it into sequences, they were padded with zero, the reason being to keep all the encoded sequences of the same length. The above step was implemented on both training and testing data – with the following parameters:
- For Padding, trunc type = 'post', padding type = 'post', out of vocab - 'oov' tok = 'OOV'.
- Using One-hot encoder from scikit-learn package to convert categorical target labels to an array of encoded integer values.

5.3 Proposed neural network architectures description

This research study was carried out while considering various sequential-based and functional-based model architectures. The baseline NN consisted of an embedding layer, which was experimented with native TensorFlow embeddings, pre-trained GloVe based embeddings, and pre-trained ELMo based embeddings, stacked at the top along with ConvNet1-D layer. LSTM layer to prepare a deep NN and BERT based embeddings were used with a three-layered dense NN for performing multi-class classification on Bloom's taxonomy concerning its cognitive domain.

5.3.1 Sequential model-based architectures

- A sequential model was defined, where the first layer is an embedding layer, to get featured vector representations, with a shape = (vocab size, embedding dim, input length = train padded.shape [1]))
- Next, a ConvNet1-D layer was defined to extract features from the sequence of integers. Additionally, an LSTM layer or Bidirectional Layer with LSTM was also defined. A'relu' activation function followed which the model can use another layer to classify based on sequence-based integers. The Bidirectional Layer was considered for experimentation in this methodology to implement a back and forward pass on encoded sequences to make sequence learning efficient.
- Then a dense layer and a final output dense layer were defined, with a specific number of 6 neurons, since six unique classes had to be classified using a 'softmax' function to perform a multi-class classification with probabilistic distributions.

5.3.2 Training specifications

Table 5 shows that trainable parameters were used while implementing the deep NN architecture. Evaluations of various experimented NN architectures and their F1 scores, accuracy and loss have been tabulated in the results section.

Table 5 Trainable parameters used for implementation of NN, and BERT based Model

Trainable parameters	Values implemented
Optimizer	Adam optimizer (NN, NN with ELMo embeddings); 'Adam' with a slow learning rate of about 1e-5 (BERT based Model);
Epochs	10
Batch size	8
Regularization	Regularization like dropouts, recurrent dropouts, and gradient clipping was also considered to prevent overfitting data and exploding gradients.
Loss function	**Categorical Cross-Entropy**, H(p, q) was chosen as the loss function and the equation is given as follows, $H(p, q)=\sum_{i=1}^{n} p_i \log q_i$ Where p_i is the truth label, qi is the 'softmax' probability for class, and n is the number of classes.

5.3.3 Functional-based model architectures

A functional-based model was defined to experiment with state-of-the-art architectures like the BERT-based model and ELMo-based contextual embeddings.

5.3.4 *BERT based model*

Bidirectional Encoder Representations from Transformers, commonly known as BERT, is a transformer-based machine learning mechanism for NLP. It is implemented on a wide variety of day-to-day complex NLP tasks.

To extract BERT based embeddings, the given input text data must be processed and converted into the following type of embeddings :

Token embeddings - Special tokens are inserted into the beginning, like < CLS > and at the end of a specific sequence, like < SEP > to depict an entire sequence.

Position embeddings depict an individual token's position in a particular sequence. Segment embeddings help distinguish embeddings of different sentences in the same sequence. The BERT language model considered for this research methodology was a small, uncased BERT model, consisting of 12 layered encoders, pre-trained with WordPiece vocabulary, and requires the use-case of all the embeddings mentioned above. A critical limitation of BERT is that the given maximum length of a specific sequence for input in BERT must be 512 tokens. Shorter sequence inputs, which have small sequences when compared with the maximum allowed input size, pad tokens < PAD > are used. If a given input text length exceeds the allowed sequence length, it must be up to the allowed length. Figure 5 shows the proposed sequential pipeline for performing Question Classification.

The pre-trained BERT uncased model layer was used using the TensorFlow's wrapper called KerasLayer, which wraps a saved model or callable object with the 'trainable' parameter set to 'True' so that it could be fine-tuned on the given dataset. Tokenization was handled via the Full Tokenizer class obtained from TensorFlow's official GitHub repository. It was implemented to build word vocabulary over the given dataset. The input text from the dataset was processed up to the given maximum length, up to 512 characters, and was added with two special tokens at the beginning and end of every certain sequence. The excess remaining length of the

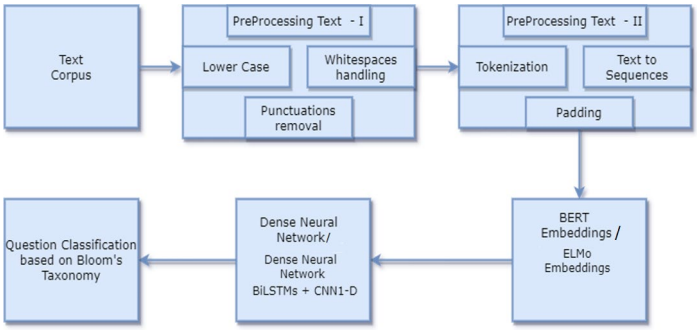


Fig. 5 Proposed sequential pipeline for performing question classification

sequence was padded with <PAD>tokens to provide attention to only the specific sequence tokens, rather than on pad tokens. All the token ids, segment ids, and mask ids (binary integers representing whether a specific sequence index is a sequence token or a padding token) were appended to the individual lists via a helper function. So that they can be mapped while fine-tuning the NN along with BERT-based embeddings extraction. This helper function was incorporated to prepare encodings for BERT based model layer's input. The labels were, as usual, encoded to the one-hot encoding of array representations. Table 5 shows the trainable parameters for the BERT model.

5.3.5 Training specifications

- A specific functional model was defined, consisting of KerasLayer wrapper for BERT Embedding, which takes token ids, mask ids, and segment ids as input for each specific sequence at a particular iteration and returns two sequence output and pooled outputs.
- The pooled output obtained was passed to the further connected feed-forward dense NN, which consisted of a few dropout layers to prevent the overfitting of data.
- The following parameters were used with given values while compiling and training the model :
- Then a normal dense layer and a final output dense layer were defined, with a total number of 6 neurons, since there are six unique classes with a 'softmax' function to implement a multi-class classification with probabilistic distributions.

The results section has tabulated evaluations of BERT-based dense NN architecture and its metrics and performance plots.

5.3.6 ELMo based embeddings

ELMo, a state-of-the-art NLP framework, is deep contextualized based word representations often used to model words' semantic characteristics and their respec-

tive context in a particular sequence. These word vectors are learned functions of the internal states of a deep bidirectional language model, thus getting forward and backward pass to do training and learning of word vectors more effective and, hence, pre-trained on a large text corpus.

This architecture uses a CNN-based character level framework to represent words of given text input into different word vectors. These vectors are later passed as inputs to the bidirectional language model's first layer. The forward pass holds information about a given the word and the context before that word. On the other hand, the backward pass contains the necessary information about the word and the corresponding context after it. This joint pair holding information from both the forward and backward pass makes an efficient learning mechanism for this model and therefore develops the intermediate word vectors and is then put into the subsequent layer. The last representation, ELMo embedding, is a total weighted sum of the given raw word vectors and the two intermediate word vectors.

For this research methodology, pre-trained ELMo embeddings were used. This pre-trained model has 1024 fixed dimensions. To access this pre-trained embedding, TensorFlow hub's 'module' class, which showcases a part of a TensorFlow graph that can be stored and later re-loaded, was used to load the ELMo pre-trained embedding module. Module's parameter, 'trainable' was set to True, so it is trainable on the custom data (Table 5).

5.3.7 Training specifications

ELMo module, once loaded by including the use-case of module object from TensorFlow hub, was set with the following parameters:

As_dict - True - For computing output dictionary of unique words and their respective unique indexes while training on the custom dataset. A custom function was designed to extract embeddings using the ELMo pre-trained module. To account for an entire sequence vector representation, the mean of given ELMo vectors of the inherent terms of a specific sequence was processed at a particular iteration during training. The function was passed as a Lambda layer to obtain embeddings using ELMo based helper function. The extracted embeddings were then passed to a ConvNet1-D layer to extract features from text sequences, and after that, the output obtained from this layer was passed over to a Bidirectional LSTM layer. The following parameters were used with given values while compiling and training the model. Then a Normal Dense layer and a final output dense layer were defined, with a maximum of 6 neurons, since six unique classes are used with a 'softmax' function to carry out a multi-class classification with probabilistic distributions. Evaluations of ELMo embeddings-based neural network architectures and their metrics are tabulated under the results section.

5.4 Computing semantic similarity between generated and manually drafted questions and retrieval of explicit recommendations of questions based on similarity.

So far, the proposed network architecture generates questions from an input text corpus and performs classification on the questions for cognitive levels. This study provides diverse questions with aspects of contextual awareness to faculties. An essential component is proposed - providing recommendations of diverse questions

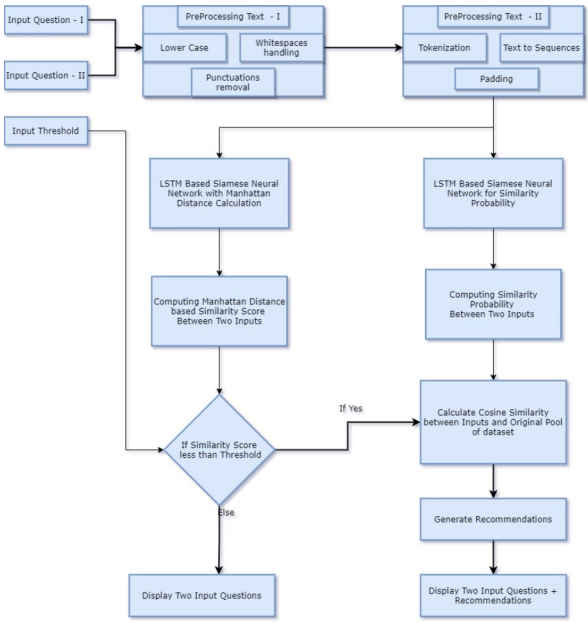


Fig. 6 End-to-end pipeline for computing semantic similarity distance metric and generating explicit recommendations based on given input threshold

per semantic similarity obtained as a Manhattan distance metric between a pair of given questions, i.e., generated questions and manually drafted questions, controlled by a threshold over a specific metric. Here, the threshold values used are 0.2, 0.3, and 0.45, respectively, for generating explicit recommendations based on the input obtained from the pipeline. Considering a scenario where a user obtains a pair of two questions from the first pipeline and these two questions are assessed based on computed semantic similarity. This computed semantic similarity between questions, i.e., generated via pipeline and questions already present in a pool manually drafted earlier, helps assess and analyze how much the generated questions from the proposed pipeline are like the questions annotated manually. If they are similar, the proposed pipeline automates the desired process of providing recommendations from the pool of questions stored in a database, based on the input threshold provided with the initial configuration. This threshold is specified concerning a similarity metric, i.e., if the similarity score between two questions is much less, indicating the given input questions are much similar. This score is less than the specified input threshold; the automated pipeline function could be called to generate recommendations of more contextually diverse questions from the pool of questions stored in a database.

The contextual similarity depends on the provided input corpus at the first pipeline component, i.e., at the Question Generation Module. This study was proposed to explore insights and analyze the efficiency of the proposed pipeline – how much the proposed pipeline is effective in generating diverse questions to input corpus. Compared to the manual drafted questions and providing recommendations of questions, which requires comparatively more time and arduous manual work. Figure 6 represents the schematic end-to-end pipeline description of this specific module.

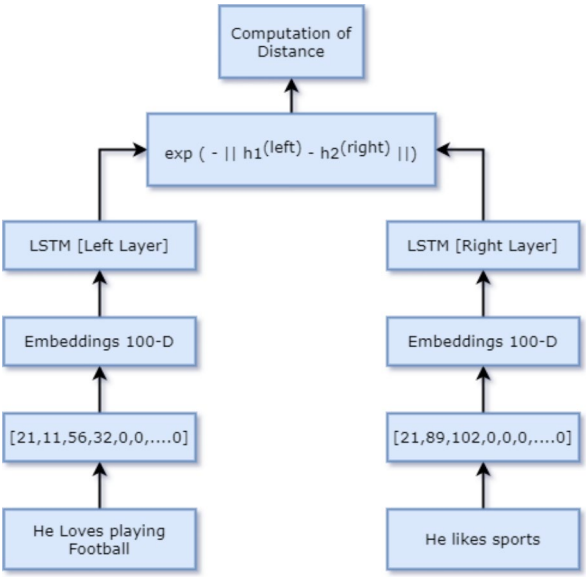


Fig. 7 Schematic representation of LSTM-based Siamese NN for computing Manhattan distance for semantic similarity

5.3.8 Working out the semantic similarity between generated questions and manually annotated questions

Mueller & Thyagarajan (2016) developed a Siamese NN with an alteration of the LSTM network for a given labeled pair of text data. Their research proposed dense NN and word-embedding vectors passed to LSTMs to assess the semantic similarity between these sentence pairs, which used a fixed-size vector to encode the fundamental definition expressed in a sentence. Figure 7 represents LSTM-based Siamese NN implementation for computing text similarity based on Manhattan distance.

This work proposes a deep learning based Siamese NN for calculating Manhattan distance, a simple metric to evaluate similarity distance between two given text inputs. The network takes in two text inputs and evaluates the similarity distance between two given text inputs. The model was made such that it gives the best generalization accuracy.

5.3.9 Dataset preparation

Two datasets were chosen for training the proposed component. The first dataset was obtained from Kaggle, a crowd-sourced platform for available datasets. In contrast, the other dataset was prepared and manually annotated by University faculty through a survey.

The first dataset used for training the model was the Quora Question Pairs dataset from Kaggle. This dataset has two sets of end-to-end questions, some questions being the same and others being different. The NN was put into training on this dataset. It has around 4,04,290 question pairs. A sample of the dataset is depicted in Table 6.

The dataset has the following parameters:

Table 6 Sample of Quora dataset used for training Siamese-based LSTM network for Manhattan distance

id	qidl	qid2	question1	question2	is_duplicate
0	1	2	What is the step-by-step guide to investing in the share market in India?	What is the step-by-step guide to investing in share market?	0
6	13	14	Should I buy Tiago?	What keeps children active and far from phones and video games?	0
19	39	40	Which is the best digital marketing institution in Bengaluru?	Which is the best digital marketing institute in Pune?	0
7	15	16	How can I be a good geologist?	What should I do to be a great geologist?	1
12	25	26	What can make Physics easy to learn?	How can you make physics easy to learn?	1
16	33	34	What does manipulation mean?	What does manipulation mean?	1

Table 7 Sample of survey dataset obtained via google forms

Passage	Answer Context
Everybody very frequently uses the term health. How do we define it? Health does not simply mean "absence of disease" or "physical fitness". It could be defined as a state of complete physical, mental, and social well-being. When people are healthy, they are more efficient at work. This increases productivity and brings economic prosperity. Health also increases the longevity of people and reduces infant and maternal mortality. When the functioning of one or more organs or systems of the body is adversely affected, characterized by the appearance of various signs and symptoms, we say that we are not healthy, i.e., we have a disease. Diseases can be broadly grouped into infectious and non-infectious. Diseases which are easily transmitted from one person to another are called infectious diseases.'	Easily transmitted from one person to another
Proteins are the most abundant biomolecules of the living system. Chief protein sources are milk, cheese, pulses, peanuts, fish, and meat. They occur in every part of the body and form the fundamental basis of the structure and functions of life. They are fundamental for the growth and maintenance of the body. The word protein is derived from the Greek word, "proteios" which means primary or of prime importance.	Greek Word

- id: Unique ID of the pair of a training set data.
- qid1, qid2: Unique ID of the given Question.
- question1: Text phrase for the first Question.
- question2: Text phrase for the second Question.
- Is_duplicate (Target label) l: would be 1; if question1 and question2 have the same meaning, else 0.

A separate dataset was prepared with the help of a survey among faculties to test the model's performance. Manual annotation was studied through a survey. This survey contained a series of comprehensive passages on various topics that were carefully chosen from school-level textbooks like NCERT for the 12th standard in the following subjects - Physics, Chemistry, History, and Biology (NCERT, 2021). Each passage was chosen in such a way, such that none of the survey takers would face any lack of domain knowledge. The survey contained four short passages (ranging from around 60–160 words each). Each passage was chosen from the subjects, two chosen answer contexts for each, and a sample (Table 7**)**.

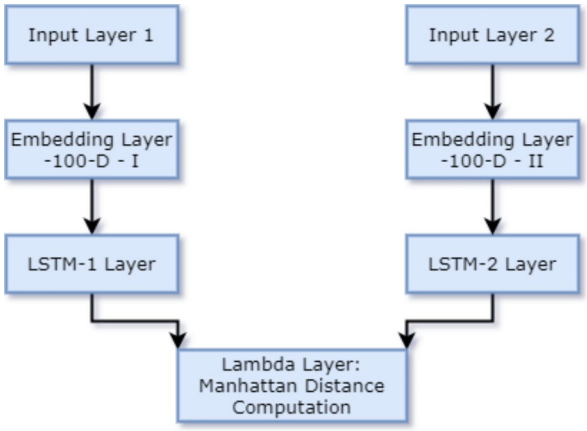


Fig. 8 NN Layers Plot with Siamese NN

These were the same passages that were passed through the present fine-tuned model mentioned in the Question Generation module, which was then used to generate questions. The survey required the end-user to read all these passages one by one, and for each passage, they were given two contextual words/phrases. They then had to frame a question for each contextual word/phrase and answer these using the knowledge gained from the passage they read. Afterward, the present testing was done on these generated questions from the model and the received questions from the survey.

5.3.10 Pre-processing phase

Before working on the core NN, all questions in the system were subject to the following pre-processing steps as described in 5.2.2 with the following parameters:

- The vocabulary size = 50,000 and maximum length = 240.
- This pre-processing was carried out with the above step with training and testing data – questions and target labels.
- Since the target column 'is duplicate' already had an integer binary value, an array of the target and the test target labels were being made using one-hot encoding.

5.3.11 Baseline neural network architecture

The authors propose a dense NN to compute Manhattan distance, which could help compute the similarity between two given text phrases.

- Manhattan distance: A custom function was defined to calculate passed sequences' negative exponential Manhattan distance as parameters, as provided in Table 8.
- Input Layer: The two input layers have an input shape equal to the maximum length of a sentence from the entire corpus for the first and second questions.

Table 8 Trainable parameters used for implementation of Siamese NN with Manhattan distance computation

Trainable Parameters	Values Implemented
Optimizer	Adam with parameter 'clipnorm'=1.25, to avoid exploding gradients with the help of gradient clipping.
Epochs	50 epochs.
Batch size	128
Regularization	Regularization like dropouts, recurrent dropouts, and gradient clipping was also considered to prevent overfitting data and exploding gradients.
Loss function	Binary Cross-Entropy. For training data, H(p, q) was chosen as the loss function, binary cross-entropy, and its equation is given as follows. $\mathrm{H(p,q)} = -\frac{1}{N}\sum_{i=1}^{n} y_i \log\left(p(y_i)\right) + (1-y_i)\log(1-(p(y_i))$ where $p(y_i)$ is the ith scalar value in the model output, y_i is the corresponding target value, and 'N' is the number of scalar values, as seen in the model output.

- Embedding Layer consists of featured vector representations of the word vocabulary extracted from the word tokenizer during the text pre-processing phase. For this, pre-trained vectors from the GloVe algorithm with fixed dimensions of 100 were used. After importing, the authors designed a custom embedding matrix by including the use-case of loaded GloVe vectors and word vocabulary extracted from the tokenizer. This returned an embedding matrix containing all the text sequences in featured vector format for training and testing both. Both questions could then pass through this embedding layer.

- LSTM Layer: LSTM Layer with 128 neurons was kept as a part of the proposed dense NN to incorporate sequential learning.
- Lambda Layer: Lambda Layer was defined to create an arbitrary expression for this model in the sequential flow after defining the above individual layers. Lambda layer was defined to pass the embedded text sequences to the Manhattan distance custom function; this layer could be used to calculate the real distance between two passed text sequences.
- Model: Finally, this layer was used to map the inputs and outputs for the NN in a sequential flow to incorporate a fully stacked NN for these inputs and outputs.

Shape = (inputs=[ques1,ques2], outputs=manhattan_distance).

- Final Network: After defining the individual layers, the layers were stacked upon one another in this dense NN, as defined in Fig. 8.

5.3.12 Training specifications

The following trainable parameters were used while implementing the deep NN architecture.

The results section has tabulated and depicted evaluations of Siamese-based dense NN architecture and its metrics and performance plots. Table 9 depicts a sample of obtained outputs as Manhattan distance similarity scores between the generated ques-

Table 9 Sample output of Manhattan distance scores obtained between generated and manually drafted questions

Generated Questions	Manually Annotated Questions	Context	Similarity Score
Why is health most efficient at work?	How does health affect our efficiency at work?	Increasing productivity and bringing prosperity	0.4432
What is the health of people more efficiently at work?	What are the outcomes of being more efficient at work due to good health?		0.4801
What is the term 'infectious disease'?	What do you mean by an infectious disease?	Diseases are transmitted easily from one person to another	0.3501
How are infectious diseases explained?	Define an infectious disease.		0.2460
According to the classical electromagnetic theory, an accelerating charged particle can do what?	According to electromagnetic theory, what happens when charged particles accelerate?	Radiations emitted in the form of electromagnetic waves	0.2080
What does the theory of an accelerating charged particle mean?	What does the classical electromagnetic theory state?		0.0461
What was the Harappans's strategy for sending expeditions?	What was the primary reason for early settling and expeditions as seen by Harappans?	Techniques for obtaining raw materials	0.4211
What was the core idea behind sending expeditions to Rajasthan?	Why did the Harappans send the expeditions to areas in Rajasthan?		0.6855
What was one feature of culture in Ganeshwar?	What was the most distinctive feature in Ganeshwar culture?	Historical Non-Harappan Pottery	0.6429
What type of artifacts are found from the Ganeshwar culture?	What different artifacts are there in Ganeshwar culture?		0.4209
Proteins can form the basis of what?	What is the key significance of proteins?	Important bio-function of life	0.1901
What are proteins the fundamental basis of?	What does a protein form under the fundamental basis of structure?		0.1750

tions from fine-tuning the transformer and manually annotated questions obtained via survey.

5.4 Generating explicit recommendations of questions

After computing the semantic similarity with the Manhattan distance metric from the two sets of questions generated from a module and manually annotated questions from the pool, a specific sub-component is being proposed to provide explicit recommendations of diverse contextual questions, provided with the fact. The Manhattan distance score is less than the given input threshold.

This study included implementing content-based recommendations using cosine similarity for generating questions (Chali et al., 2009; Jayakodi et al., 2016; Pal et al., 2022). The content used to compare given questions was analyzed using a similar approach mentioned in the above sub-component, i.e., using a Siamese-based LSTM NN but without any explicit metric like Manhattan distance. The authors propose a deep learning based Siamese NN for predicting only the probability for given inputs, whether the input pair of questions are similar or not, i.e., with a similar approach to the binary classification task. The returned pair of probabilities helped understand

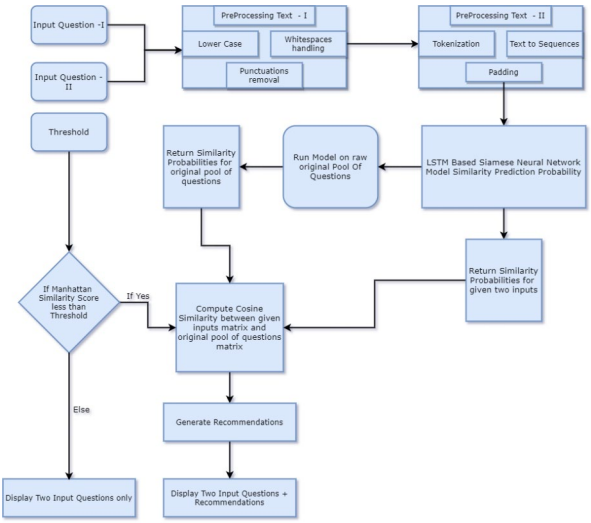


Fig. 9 Pipeline for content-based recommendations over given input

the relation between given inputs. Once a model gets trained on a particular input dataset, the trained weights of that NN are extracted and stored separately. They are used to compute predictions of a similar pair of questions provided as input from the test data.

The extracted model weights were later used to compute probabilities on the original pool of manually annotated datasets. Later, the automated pipeline computes the cosine similarity between the returned predictions of two given inputs and return predictions of a manually annotated pool of questions stored in a database, generating recommendations based on the cosine similarity scores. Figure 9 describes the entire pipeline processing for content-based recommendations retrieval using threshold input, Manhattan distance metric retrieved from the first sub-component in this pipeline, and input questions.

5.4.1 Dataset preparation

The survey dataset consists of a pool of manually annotated questions via a survey. It is a series of comprehensive passages on several topics that were carefully chosen from school-level textbooks NCERT Based books for 12th standard. Therefore, this research methodology used a package called NLPAug, which is used to implement textual data augmentations using unsupervised and supervised approaches to generate more textual data for training purposes. NLPAug provides three types of different augmentations that can be carried out to generate textual augmentations: Character level augmentations, Word level augmentations, and sentence level augmentation. This package also includes handling non-contextual embeddings like GloVe, word-2vec, or contextual embeddings like BERT, Roberta, for data augmentation. NLPAug provides specific methods like random deletion of some words, random insertion of words, shuffling, and synonym replacement of words in sequences for generating more diverse data. This study included a word-level synonym replacement technique via pre-trained word embeddings generated using an unsupervised GloVe algorithm.

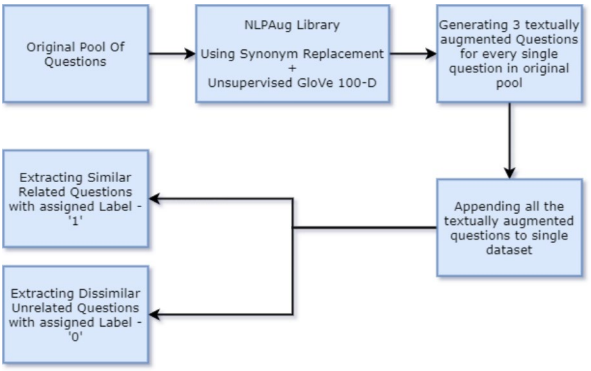


Fig. 10 Data augmentation pipeline using NLPAug and extraction of other datasets

Table 10 Textually data augmented questions based on original dataset via NLPAug library

Original Question	Data Augmented Question
Why did the Harappans send the expeditions to diverse areas in Rajasthan and South India?	Why soon the Harappans send expeditions to enough Rajasthan and South India areas?
	Why did the Harappans send expeditions time several in Rajasthan and South India?
Why Harappan expeditions established communication with local communities?	Why Harappan expeditions established communication with local churches?
	Why Harappan expeditions established communication with municipal communities?
Describe the model of an atom proposed by Rutherford	Describe the working of an atom argued by Rutherford.
	Describe the model-new with atom proposed by Rutherford.
What is the difference between the structure of an atom comprising nucleus and revolving electrons and sun planet system?	What is the difference between the structure of an atom comprising epidermal and revolving electrons and the sun planet system?
	What are the terms between the structure of atom comprising nucleus and revolving electrons on sun planet uses?
When people r healthy n more efficient when at work	Why is health almost efficient at work?
	What does health do to bring efficiency at work?
How do infectious diseases spread?	How do infectious diseases bring?
	How may infectious diseases identify?

This technique replaces ‘n’ number words with synonyms (different word embeddings obtained via pre-trained GloVe algorithm, close to those words) to retrieve a sentence with the same meaning but different words. Figure 10 describes the data augmentation process by including the NLPAug package’s use case.

The original dataset only consists of around 308 questions from the survey. A manual script was designed to perform textual data augmentation on this original dataset using the NLPAug package in such a way that it could generate up to three different augmented questions based on every single Question from the original dataset. Hence, around 924 (308 * 3) augmented questions were generated. Table 10 describes a sample of textually augmented data questions obtained by running a custom script over the original annotated dataset.

This augmented dataset was merged with the survey dataset later, and this new merged dataset was used to annotate and make two different new datasets manually. The first new dataset consisted of a pair of contextually related questions and was

Table 11 Original pool of questions

id	question_title	Topic
8	How did the Harappans procure raw material for craft production?	History
10	What was the primary or main reason for settlements and expeditions observed by the Harappans?	History
109	What are the various parts of an atom	Physics
177	Define radioactive decay as constant	Physics
223	How are infectious diseases identified?	Biology
224	Where is the word protein derived from	Biology

labeled '1'. On the other hand, the second new dataset was being formed such that the pair of questions were contextually unrelated and assigned a label '0'.

Overall, there were three different datasets :

- The merged dataset consists of all unique, augmented, and original questions with unique ids to identify them.
- Similar Related Dataset consisted of a pair of contextually similar questions and was assigned label '1'.
- Dissimilar Dataset consists of a pair of questions that were not similar contextually and were assigned label '0'.

The similar related and dissimilar related datasets did have unique question ids, which were finally taken from the original merged dataset. The original pool of questions with unique ids was stored in a dataset and later incorporated to provide recommendations based on the automated pipeline. The similar related and dissimilar datasets were merged, shuffled, and then split using a train test split from the scikit-learn package to train the Siamese-based LSTM NN. Table 11 depicts a portion of the original pool of questions and its concerned Topic Area. Table 12 depicts a portion of a related and unrelated dataset with assigned labels as '1' and '0'.

5.4.2 Pre-processing phase

Before working on the core NN, all questions in the system went through the following pre-processing steps as described in 5.2.2 with the following parameters:

- The vocabulary size = 50,000 and maximum length = 240.
- This pre-processing was done with the above step with training and testing data – questions and target labels.
- Since the target column 'is duplicate' already had an integer binary value, an array of the target and the test target labels were being made using one-hot encoding.

5.4.3 Baseline neural network architecture

This work proposes a Siamese-based dense NN to compute the first similarity and then extract the trained model weights to compute predictions on custom pair of inputs. Then compute cosine similarity scores to get content-based recommendations

Table 12 A sample of similar-related (One with Label '1') and dissimilar-related (One with Label '0') Datasets

first_ques	second_ques	first_ques_id	second_ques_id	Label	Dataset
Why did the Harappans send their diverse expeditions to different areas in Rajasthan and South India?	Evidence relating to the Ganeshwar-Jodhpura culture has been found in which place? What are they famous for?	0	36	1	Similar related dataset
What are the end results of being much more efficient at work because of good health?	What are the true benefits of staying healthy regularly?	195	189	1	
Obtain an expression for the frequency of radiation emitted when a hydrogen atom de-excites from level n to level (n–1). For large n, show that this frequency equals the classical frequency of revolution of the electron in orbit.	What are the parts of an atom	156	109	1	
What are the techniques used for obtaining raw materials in the Harappan civilization? What distinct factors does this depend on?	According to the classical electromagnetic theory, in what form does the accelerating charged particle emit radiation?	2	141	0	Dis-similar related datasets
What were the confusions in their mind of Cunningham while studying Harappan civilization?	What does health increase and bring?	59	200	0	
What does the theory of accelerating charged particles posit?	What are the various routes by which transmission of the human immunodeficiency virus takes place?	130	271	0	

of the questions whose scores are relevant to the past pair of inputs. Figure 11 depicts the proposed NN-based pipeline for computing the probability prediction relation between two given text inputs.

Two different networks have been experimented with - Siamese NN with LSTM and Siamese NN with Bidirectional LSTM (Table 8).

- **Input Layer**: The two input layers have an input shape equal to the maximum length of a sentence from the entire corpus for the two questions.
- **Embedding Layer**: The embedding layer consists of featured vector representations of the word vocabulary used from the word tokenizer during the text preprocessing phase. Hence, pre-trained vectors from the GloVe algorithm with fixed dimensions of 100 were used. After importing, the authors designed a custom embedding matrix by including the use-case of loaded GloVe vectors and word vocabulary extracted from the tokenizer. This returned an embedding matrix containing all the text sequences in featured vector format from training and test data. Both questions could pass through this embedding layer.
- **LSTM Layer**: The LSTM Layer with 128 neurons was a part of the dense NN to incorporate sequential learning.
- **Dot Layer**: After passing the individual text inputs to LSTM, a custom layer was designed to calculate the dot product between these two pairs of text inputs, taken from the above LSTM layer being flattened out.
- **Dense Layer**: The dot product resulted in a concatenation of flattened vectors, which was passed to a dense layer with an output equal to '1', computing the probability of whether the two pairs are similar. This layer was provided with a 'sigmoid' activation function.
- **Model**: Lastly, this layer was used to map the inputs and outputs for the NN in a functional flow to incorporate a fully stacked NN with these inputs and outputs.

Shape = (inputs=[ques1,ques2], outputs=output_dense).

- **Final Network**: After defining the individual layers, the layers were stacked upon one another in this dense NN, as defined in Fig. 12.

5.4.4 Training specifications

The trainable parameters used while implementing the deep NN architecture are shown in Table 13

5.4.5 Recommendation module

After saving the trained model, its weights were extracted and saved separately. A custom function was developed to get recommendations of questions from the initial pool of questions stored in the database. The two custom inputs were first converted to sequences and were padded. This step was done using the same tokenizer and

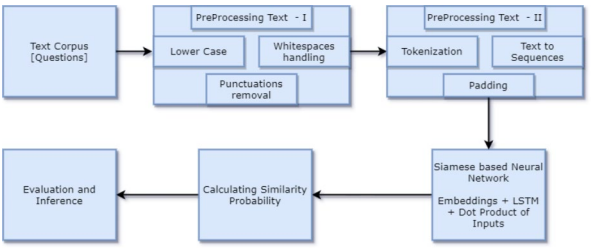


Fig. 11 Proposed pipeline for predicting similarity probability relation between text inputs

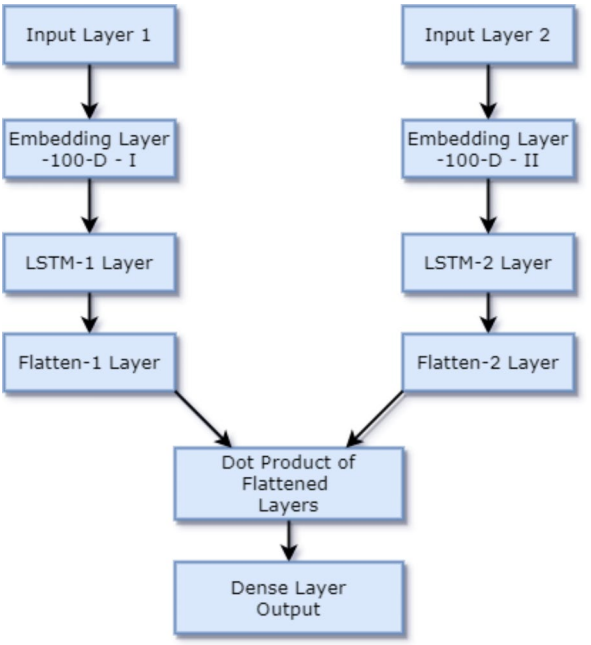


Fig. 12 NN layers plot with Siamese NN.

Table 13 Trainable parameters used for implementation of Siamese NN.

Trainable Parameters	Values Implemented
Optimizer	Adam with parameter 'clipnorm'=1.25, to avoid exploding gradients with gradient clipping.
Epochs	5 epochs.
Batch size	16
Regularization	Regularization like dropouts, recurrent dropouts, and gradient clipping was also considered to avoid overfitting data and exploding gradients.
Loss function	Binary Cross-Entropy. For training data, H(p, q) was taken as the loss function, binary cross-entropy, and its equation is given as follows. $\mathrm{H(p,q)} = -\frac{1}{N}\sum_{i=1}^{n} y_i \log\left(p(\,y_i\,)\right) + (1-y_i)\log(1-(p\,(y_i))$ Where $p(y_i)$ is the ith scalar value in the model output, y_i is the corresponding target value, and N is the number of scalar values in the model output.

keeping the state similar, which is used to train the model. The saved model weights help predict these two given inputs and return a matrix of probabilities. Without data

Table 14 Explicit recommendations output based on given input threshold obtained via the original pool of questions

Threshold	Question 1	Question 2	Recommended Questions based on threshold
0.2	What does the theory of an accelerating charged particle imply?	What does the classical electromagnetic theory state?	1. According to electromagnetic theory what does an accelerating charged particle do? 2. What does the model of atom proposed by Rutherford assume? 3. Describe the model of an atom proposed by Rutherford.
	Proteins form the basis of what?	What is the significance of proteins?	1. Why is health more efficient at work? 2. What does health increase and bring? 3. Protein is derived from which word?
0.3	How are infectious diseases defined?	Define infectious disease.	1. How do infectious diseases spread? 2. What are infectious diseases? 3. Define infectious disease.
	According to classical electromagnetic theory, an accelerating charged particle does what?	According to electromagnetic theory what happens when a charged particle accelerates?	1. What is the Rutherford model of atom? 2. What does an accelerating charged particle emit? 3. According to classical theory, accelerating charged particles emits what?
0.45	What are proteins the fundamental basis of?	What does protein form along with the fundamental basis of structure?	1. What do proteins form in the body? 2. How do proteins help our body? 3. What are proteins?
	What was the Harappans strategy of sending expeditions to?	What was the primary reason for settlements and expeditions as seen from Harrapans?	1. How did the Harappans procure raw material for craft production? 2. Why did the Harappans settle in specific regions? 3. What was the Harappans strategy of sending expeditions to?

augmentation, the original dataset was converted into a dictionary with its respective unique ids of the discrete questions. These questions, too, were first converted to sequences and were padded. This step was done using a similar tokenizer and in the same state, it was used for model training. The saved model weights were being used to predict the questions of this dictionary to return another matrix of probabilities. Cosine similarity was computed among these two sets of matrices from custom inputs and one from the original pool of questions. A data frame consisted of the most pertinent question ids with relevant cosine similarity scores appropriate to the custom input questions.

The question ids were cross-referenced to their respective questions from the original dataset and recommended to the user. The above-mentioned automated pipeline works well regarding the threshold range. However, users can also explicitly call for question recommendations without the requirement for a threshold by just providing two input sets of questions. This ensures that recommendations can generate contextually diverse questions with every input per user's convenience and create questions from the original pool of questions without any hassle, saving time and effort. Table 14 shows the sample of content-based recommendations constructed on a custom input threshold.

6 Results and discussion

6.1 Question generation module

To assess the questions generated from the current model, certain metrics were used. For this method, the generated ones were compared with the augmented and survey data set discussed in 3.2.2. The model can generate two semantically different questions for each specified condition and answer. For every one of these two questions, two were taken in the present augmented and survey dataset, which answered the same context and was also used for evaluation (Alami et al., 2021; Lamsiyah et al., 2021; Khurana & Bhatnagar, 2022). Three metrics were used for this purpose. It was first being the ROUGE score and calculating the F1 score. The second and third methods used were primarily to compute the semantic similarity of these two sentences. It was centered on two different sentence transformer pre-trained models - 'sentence-transformers/stsb-roberta-large' and 'sentence-transformers/bert-base-nli-mean-tokens'. These two methods use pre-trained embeddings to compute the cosine similarity of the two sentences which is taken as input, giving a score in the range of 0 and 1. The outcomes are displayed for four subjects, having two sets of questions each, and are showcased in Table 15.

6.2 Question classification module

Different architectures were trained on the given dataset and all the observations are provided in Table 16.

6.3 Calculating semantic similarity between questions via Manhattan distance metric

The model trained on the Quora questions dataset first with the following results of validation accuracy and loss is 81.79% and 0.4784, respectively. Evaluation is also depicted in Figs. 15 and 16, respectively.

This study is representative of a pair of noted observations that the network evaluated while computing similarities between different encoded text phrases. This indicates how the similarity score pans out for various questions, both generated and manually interpreted, exhibiting how far these featured word representations lie concerning one another.

6.4 Generation of explicit recommendations based on threshold and similarity metric

This module had two specific sub-components. One was a Siamese-based LSTM architecture, and another used a cosine similarity metric to generate recommendations based on weights extracted from the first sub-component. The first sub-component, the Siamese-based NN, was experimented with both LSTM and Bidirectional LSTM layer with the following results: Siamese+LSTM+GloVe 100-D (Accuracy=96.10%; Loss=0.98; F1 Score=0.96), Siamese+BiLSTMs+GloVe 100-D

Table 15 Evaluation of the generated questions with augmented questions using various metrics

Subject	Generated	Reference	ROUGE 1	stsb-roberta-large	bert-base-nli-mean-tokens
Chemistry	From what Greek word does "proteios" originate?	Mention the Greek word from which the second word 'Protein' has been known.	0.21	0.51	0.78
	From what Greek word does "proteios" come?	Proteios says a word of which origin?	0.14	0.74	0.65
	Proteins form the fundamental basis of what?	Does proteins form the fundamental basis for functions such as experience?	0.47	0.50	0.92
	Proteins are the fundamental basis of structure and what?	What does protein form along with fundamental possible first structure?	0.11	0.58	0.80
Biology	Why is health important to the human body?	What think health and capability at work bring about?	0.12	0.41	0.76
	Why is health important to the body?	What are the many benefits of being healthy?	0.13	0.63	0.71
	What is infectious disease?	What so infectious diseases?	0.50	0.94	0.97
	What is one definition of infectious diseases?	How does infectious diseases bring?	0.16	0.75	0.90
Physics	What is the electromagnetic theory of an accelerating charged particle?	What only and gradually charged particle do according to classic eletromagnetic theory?	0.28	0.80	0.73
	What is the electromagnetic theory of an accelerating charge?	According find electromagnetic theory what happens weeks a charged particle accelerates?	0.20	0.85	0.86
	The model of the atom is proposed by Rutherford?	What does Rutherford model of atom do?	0.37	0.84	0.96
	What is the model of the atom?	What is Rutherford's model of atom composed?	0.61	0.77	0.92
History	What was the Harappans responsible for doing?	Why did the Harappans settle in proper regions?	0.27	0.65	0.63
	What was the Harappans responsible for purchasing raw materials?	What were the follow innovative adopted by the Harrapans on source different raw materials?	0.37	0.75	0.81
	What is the hallmark of the Ganeshwar-Jodhpura culture?	What is distinct about Ganeshwar-Jodhpura culture?	0.61	0.94	0.91
	What uniqueness of the Ganeshwar-Jodhpura culture is evident?	What is distinct about Ganeshwar-Jodhpura culture?	0.42	0.97	0.87

Table 16 Evaluation of different NN architectures for classification of the cognitive domain of the Bloom's taxonomy

Model	Accuracy	Loss	F1 Score
Convnet1-D+Bi LSTM+256D	0.7647	1.19	0.76
Convnet1-D+Bi LSTM+GloVe – 100D	0.7353	1.2	0.74
Convnet1-D+Bi LSTM+GloVe – 300D	0.7647	1.13	0.76
BERT+Dense NN	0.811	0.86	0.81
ELMo+BiLSTMs	0.7941	0.83	0.79
ELMo+ConvNet1-D+BiLSTMs	0.7598	1.4	0.76

The following dense NN expressed the top performance:

BERT based embeddings with a 2-layered dense NN using Adam optimizer with a slow learning rate of about 1e-5. Figures 13 and 14 show the training vs. validation accuracy for BERT-based dense network and training vs. validation loss for BERT-based dense network. The results obtained were validation accuracy of 81.1%, validation loss of 0.86, and a F1 score of 0.81

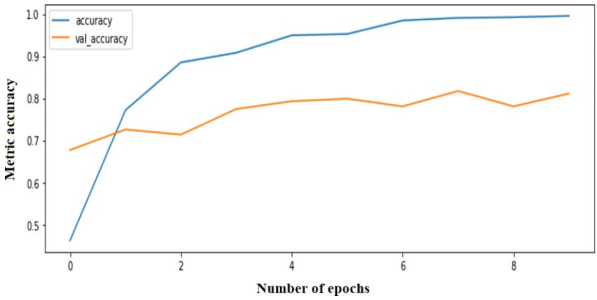


Fig. 13 Training vs. Validation Accuracy for BERT based dense Network

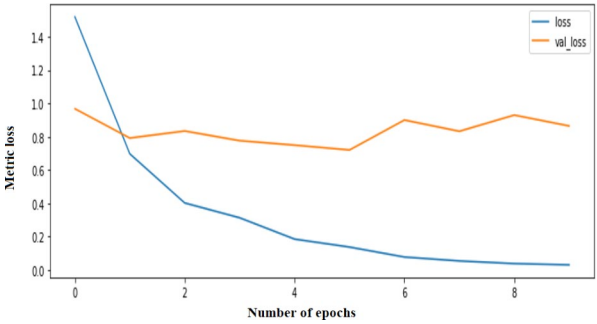


Fig. 14 Training vs. Validation Loss for BERT based dense Network

(Accuracy=94.61%; Loss=0.97; F1 Score=0.95). Evaluation is also depicted in Figs. 17 and 18, respectively.

After the model's training, its weights were used to predict the test data. Then the returned probabilities were used to compute cosine similarity against another pair of probabilities from passing a pair of input questions whose similarity metric is lesser than the input threshold, meaning that the passed input questions were similar. After extracting cosine similarity scores, the unique question ids were cross-referenced against their titles and displayed according to the pair of questions input passed.

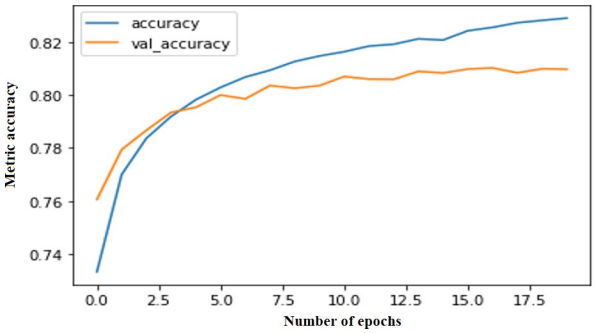


Fig. 15 Training vs. Validation Accuracy for Siamese-LSTM NN.

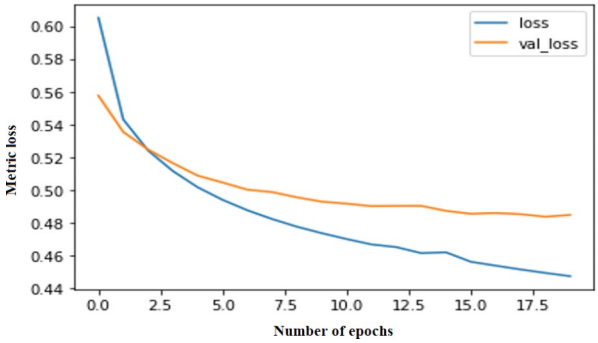


Fig. 16 Training vs. validation loss for Siamese-LSTM NN.

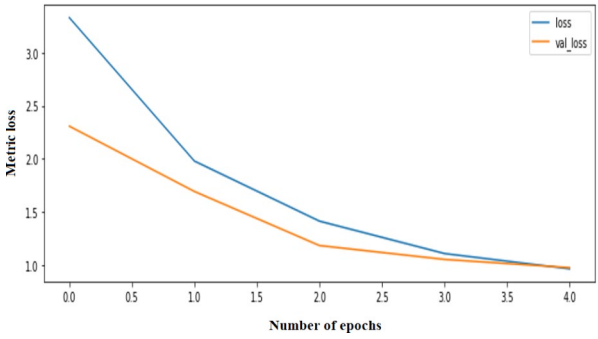


Fig. 17 Training vs. Validation loss for Siamese-LSTM NN.

A sample of recommendations was obtained while testing with different threshold scores and is summarized in the next section.

7 Conclusions and scope of future work

The present research produced an effectual pipeline for generating questions and classifying those questions built on the Bloom's taxonomy sub-domain, the cognitive level. This work provides explicit recommendations for context-based questions. The generation module also showed an ability to generate questions of a diverse range

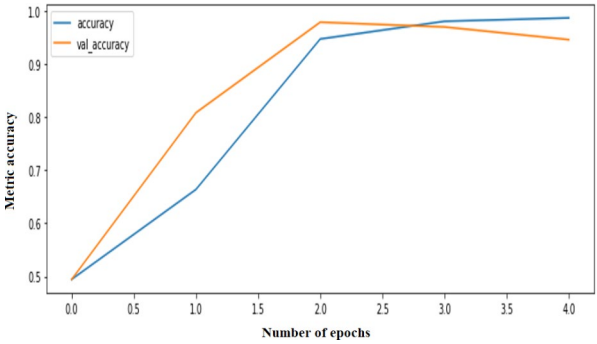


Fig. 18 Training vs. Validation accuracy for Siamese-LSTM NN.

with reliable results. The Siamese NN is implemented in the second pipeline for calculating semantic similarity distance metrics. The generation of explicit recommendations also revealed the generated questions. The manually annotated ones were of diverse classes, with the pipeline performing quite well from end to end.

Various datasets were considered for every sub-component of the presented pipelines to examine the strength of the suggested model. Several metrics were used to validate and prove the efficacy of the suggested pipeline methodology. It is concluded that the design of the suggested pipeline effectively addresses the difficulty of question generation and the Question classification as per the distinct levels of cognitive domains of the Bloom's taxonomy, along with detailed questions recommendations. This end-to-end pipeline promises an effective solution for faculties to directly get a diverse range of questions with just an input of a text corpus, obtaining generated questions directly classified under the Bloom's taxonomy for a cognitive domain for a productive inspection of the assessment levels of the questions, explicit recommendations of questions built on context. This complete pipeline promises to save extra time, rigor and effort and comprehensively provides solutions.

There are no remarkably high computational resources for working on state-of-the-art architectures like sentence-transformers and deep contextual topic modeling. Also, the data taken from the SQuAD dataset could have been increased from 3000 entries, depending on the computational power. Also, different versions of the Bloom's taxonomy were introduced in the past decades. Three taxonomies were introduced, which differed slightly. Although the focus of this work has been on the original version, one limitation is that the cognitive domains require experimentation.

Subsequently, it is hoped to investigate and experiment with different contextual passages for question generation, i.e., subjects like computer programming and mathematics. This could help to extract contextual information, using sequence to sequence learning. It would also be interesting to incorporate language models to obtain better results and evaluate using different models. Implementing other state-of-the-art architectures like sentence-transformers and deep contextual topic modeling could also provide more insights about a text corpus and lay out an idea for which topic the model can focus more on. The development of an auto-evaluation tool for rapid assessments can be considered with the aid of Siamese NN. It is explored how much a student's answer is semantically like a manually prepared answer key with

the unification of the OCR support, which can help us provide rapid analysis and assessments of a student's examination performance.

Acknowledgements The authors are grateful to the SRM Institute of Science and Technology, Kattankulathur Campus, Chennai, for supplying the required research facility.

Credit Authorship Contribution Statement **Harsh Sharma**: Conceptualization, Formal analysis, Methodology, Investigation, Original Draft-writing, **Rohan Mathur**: Conceptualization, Formal analysis, Methodology, Investigation, Original Draft-writing, **Tejas Chintala**: Conceptualization, Formal analysis, Methodology, Investigation, Original Draft-writing, **Samiappan Dhanalakshmi**: Investigation, Formal analysis, Methodology, Data visualization, Data validation, Writing - review & editing, Supervision, **Ramalingam Senthil**: Investigation, Methodology, Data curation, Data validation, Writing - review & editing.

Funding information This work received no specific grant from any funding agency in public, commercial or not-for-profit sectors.

Data Availability All the data are used in the article.

Declarations

Compliance with ethical standards Ethical clearance is not applicable.

Conflict of interest The authors declare no conflicts of interest.

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